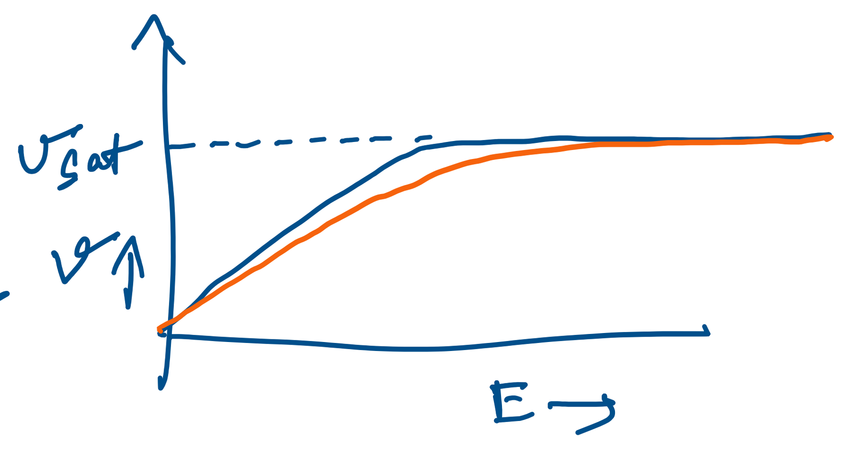
Short channel effects

Thursday, August 7, 2025 9:50 PM

Velocity Saturation

$I = \mu_n C_{ox}(V_{GS} - V_T - V)\,v$ ~ from the gradual channel approximation



$$v_n = \frac{\mu_n E_{Cn}}{1 + \frac{E}{E_{Cn}}} \quad \text{Approximation}$$

$$E_{Cn} = \frac{v_{sat} \times 2}{\mu_n}$$

$$I_D\left(1 + \frac{1}{E_C}\frac{dV}{dx}\right) = \mu_n C_{ox}(V_{GS} - V_T - V)\,W\frac{dV}{dx}$$

$$\int_0^L I_D\,dx + I_D\int_0^{V_{DS}} \frac{1}{E_C}\,dV = \int_0^{V_{DS}} \mu_n C_{ox} W (V_{GS} - V_T - V)\,dV$$

Linear region

$$I_D = \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)\left[2(V_{GS} - V_{Tn})V_{DS} - V_{DS}^2\right] \times \boxed{\frac{E_{Cn}L}{E_{Cn}L + V_{DS}}} \quad (1)$$

factor ≈ 1 when $E_C L >> V_{DS}$ which is true for long channel.

For saturation

$$I_{DSAT} = W v_{sat} Q_I$$
$$= W v_{SAT}(V_{GS} - V_T - V_{DSAT})\,C_{ox}$$

$$\boxed{v_{SAT} = \frac{\mu_n E_{Cn}}{2}}$$

$$I_{DSAT} = \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)E_{Cn}L\,(V_{GS} - V_T - V_{DSAT}) \quad (2)$$

For continuity at $V_{DS} = V_{DSAT}$, $I_D = I_{DSAT}$

$$\cancel{\frac{\mu_n C_{ox}}{2}}\left(\cancel{\frac{W}{L}}\right)\frac{\cancel{E_{Cn}}L}{E_{Cn}L + V_{DSAT}}\left(2(V_{GS} - V_{Tn})V_{DSAT} - V_{DSAT}^2\right)$$

$$= \cancel{\frac{\mu_n C_{ox}}{2}}\left(\cancel{\frac{W}{L}}\right)\cancel{E_n}L\,(V_{GS} - V_{Tn} - V_{DSAT})$$

$$2(V_{GS} - V_{Tn})V_{DSAT} - \cancel{V_{DSAT}^2} = E_{Cn}L(V_{GS} - V_{Tn}) - E_{Cn}L V_{DSAT} + (V_{GS} - V_{Tn})V_{DSAT} - \cancel{V_{DSAT}^2}$$

$$V_{DSAT} = \frac{E_C L(V_{GS} - V_{Tn})}{E_C L + V_{GS} - V_{Tn}}$$

$$\Rightarrow I_{DSAT} = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{Tn})^2 \boxed{\frac{E_{Cn}L}{E_{Cn}L + V_{GS} - V_{Tn}}}$$

≈ 1 for long channel

Unified model for nMOS

$I_D = 0 \quad V_{GS} < V_{Tn}$ } cutoff

$$= \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)\left[2(V_{GS} - V_{Tn})V_{DS} - V_{DS}^2\right]\frac{E_{Cn}L}{E_{Cn}L + V_{DS}} \times (1 + \lambda V_{DS})$$

when $V_{GS} > V_{Tn}$,

$$V_{DS} < \frac{E_{Cn}L(V_{GS} - V_{Tn})}{E_{Cn}L + V_{GS} - V_{Tn}}$$

} Linear

$$= \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)(V_{GS} - V_{Tn})^2\frac{E_{Cn}L}{E_{Cn}L + V_{GS} - V_{Tn}} \times (1 + \lambda V_{DS})$$

when $V_{GS} \geq V_{Tn}$,

$$V_{DS} \geq \frac{E_{Cn}L(V_{GS} - V_{Tn})}{E_{Cn}L + V_{GS} - V_{Tn}}$$

} Saturation

HW: Derive the unified model for PMOS.